Sandbox Mini-App Approach

CodeFixPanda_Kotlin · senior Android interview preparation · 2026-11-11

Purpose

The real app is large. To absorb it, each concept from the code walkthrough gets a tiny, runnable example in a separate sandbox Activity, `MainActivity2`. The real code stays untouched, and the sandbox can be thrown away.

Branches

Branch	Holds	Merge?
`claude/festive-mayer-purx3h`	The real code, plus the docs in `_meta/docs`: the conventions PDF and this report	Normal rules
`sandbox/walkthrough-mini-app`	`MainActivity2.kt` and a manifest that launches it	**Never**

The sandbox branch was created from the first branch, so it contains everything above plus the sandbox changes. All sandbox work goes on the sandbox branch.

What the sandbox branch changes

- **New file** `app/src/main/java/io/github/genesjc/codefixpanda/MainActivity2.kt`, the mini app.
- **Edited** `app/src/main/AndroidManifest.xml`. The launcher `<intent-filter>` moved from `MainActivity` to `MainActivity2`. `MainActivity` is otherwise unchanged.
- `CodeFixPandaApplication` still runs for both Activities, so Koin is already started and the sandbox can use the real dependency injection setup.

Switching between the sandbox and the real app

Move this block between the two `<activity>` elements in the manifest, then rebuild:

```
<intent-filter>
    <action android:name="android.intent.action.MAIN" />
    <category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
```

Under `MainActivity` the real app runs. Under `MainActivity2` the sandbox runs. A comment in the manifest marks the sandbox entry.

Working method for each chunk

1. Walk one chunk of real code, following the conventions PDF: 3 to 8 lines, explanation first, then the link and snippet last.
2. Where it helps, add a minimal example of the same concept to `MainActivity2.kt` on the sandbox branch. It should be small, with the key line commented.
3. Run it and compare what happens with what the real code does.

Progress

Chunk	Real code	Concept	Sandbox demo
1	`MainActivity.onCreate`	Activity, `enableEdgeToEdge`, `setContent`	Not yet
2	`CodeFixPandaApplication.onCreate`	Koin starts once per process, and a `single` is built lazily	Written, not yet run

Next: Chunk 3, `CodeFixPandaApp()` in `ui/CodeFixPandaApp.kt`, the root composable that `setContent` calls.

The Chunk 2 demo

It registers a tiny Koin module and logs under the tag `Mini`. The expected Logcat order is `module registered, Greeter not built yet`, then `Greeter constructed`. The second line appears only when `setContent` first reads the injected value. That shows `modules(appModule)` registers definitions without building them.

Known limits

- **Not compiled yet.** The container used for this session has no Android SDK, so the demo has not been built or run. The manifest is valid XML. The expected Logcat order is what the code should produce, not an observed result.
- The demo uses Koin 4.2.2 APIs. Calling `loadModules` again on an Activity recreation overrides the earlier definition, which Koin 4 allows by default.
- Because the manifest is edited to switch Activities, remember to move the launcher filter back before running the real app.

For another conversation joining this work

1. Read `_meta/docs/code-walkthrough-conventions.pdf` on `claude/festive-mayer-purx3h`.
2. Check out `sandbox/walkthrough-mini-app` for the sandbox files.
3. Follow the conventions. Do not merge the sandbox branch or move its changes onto the real branch.